$$N_c = 2.8 \times 10^{25} \left(\frac{T}{300}\right)^{3/2} m^{-3}$$

7.5.2 Hole Concentration in the Valence Band

An expression similar to (7.13) for the number of holes per unit volume in the energy range E and $E + dE$ can be written as

$$dp = D(E)[1 - f(E)]dE \tag{7.22}$$

where we have replaced $f(E)$ by $[1-f(E)]$ which represents the probability of an energy state E not to be occupied by an electron, i.e., the probability of finding a hole in the energy state E. Now

$$1-f(E) = 1 - \frac{1}{1+\exp\left(\frac{E-E_F}{kT}\right)} = \frac{\exp\left(\frac{E-E_F}{kT}\right)}{1+\exp\left(\frac{E-E_F}{kT}\right)}$$

In the valence band, since $E<E_F$ the exponential term in the denominator may be neglected in comparison to unity. Thus, we get

$$1 - f(E) \cong \exp\left(\frac{E-E_F}{kT}\right) \tag{7.23}$$

It follows that the probability of finding holes decreases exponentially with increase in depth into the valence band. Also, the kinetic energy of a hole in the energy state E in the valence band is (E_v-E). Therefore, the density of states per unit volume in the valence band can be written as

$$D(E) = \frac{4\pi}{h^3}\left(2m_p^*\right)^{3/2}(E_v - E)^{1/2} \tag{7.24}$$

where m_p^* is the effective mass of a hole in the valence band. Using Eqs. (7.23) and (7.24) in Eq. (7.22) and integrating from $E = -\infty$ to $E = E_v$, we obtain the hole concentration in the valence band as

$$p = \frac{4\pi}{h^3}\left(2m_p^*\right)^{3/2}\int_{-\infty}^{E_v}(E_v - E)^{1/2}\exp\left(\frac{E-E_F}{kT}\right)dE$$

$$= \frac{4\pi}{h^3}\left(2m_p^*\right)^{3/2}\exp\left(\frac{E_v-E_F}{kT}\right)\int_{-\infty}^{E_v}(E_v - E)^{1/2}\exp\left(\frac{E-E_v}{kT}\right)dE$$

Let $$\frac{E_v - E}{kT} = x$$